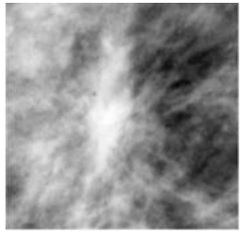
INPUT

Reference-derived ROI crop and trimap seeds
(Region of interest containing the candidate lesion)



LAYER 1: TEXTURE ENCODING

Multi-scale CLAHE Enhancement *(8x8 + 16x16 tile grids)*

LIPW / LBPW *(Local texture encoding)*

Multi-scale Local-Neighborhoods *(3x3 + 5x5 + 7x7)*

Six-channel texture representation *(standardised)*

LAYER 2: TEMPLATE MATCHING

K-means templates T_k *(5×5×6 patches of seed pixels, training set; K = 8)*

Signed template matching *(cosine with the template sign pattern, rescaled to [0, 1])*

K response maps s_k *(one similarity map per template, in [0, 1])*

LAYER 3: SPATIAL/CONTEXT AGGREGATION

Normalised 7×7 local pooling *(responses summed over a 7×7 window, divided by the sum over all K)*

K pooled responses h_k *(per-pixel profile that sums to 1)*

LAYER 4: PIXEL-WISE PROBABILITY ESTIMATION

K pooled responses h_k *(contextual representation from Layer 3)*

Local mean intensity *(5×5, 9×9 and 15×15 means of the ROI)*

RBF-SVM classifier *(balanced classes, trained on definite seed pixels only)*

Pixel-wise mass probability P *(Platt-calibrated, in [0, 1])*

LAYER 5: ALPHA REFINEMENT

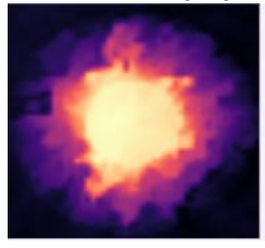
Hybrid prior
$\alpha_0 = 0.9\,P + 0.1\,I_{norm}$
(I_{norm}: ROI intensity, normalised)

Seed-constrained matting solve *(edge-aware propagation with the matting Laplacian of the ROI image; seeds held fixed)*

Refined Alpha Map ($Continuous\ \alpha \in [0, 1]$)

OUTPUT

Refined Alpha Map
Continuous $\alpha \in [0, 1]$



Thresholding $\alpha \geq 0.5$
Binary Mass Segmentation